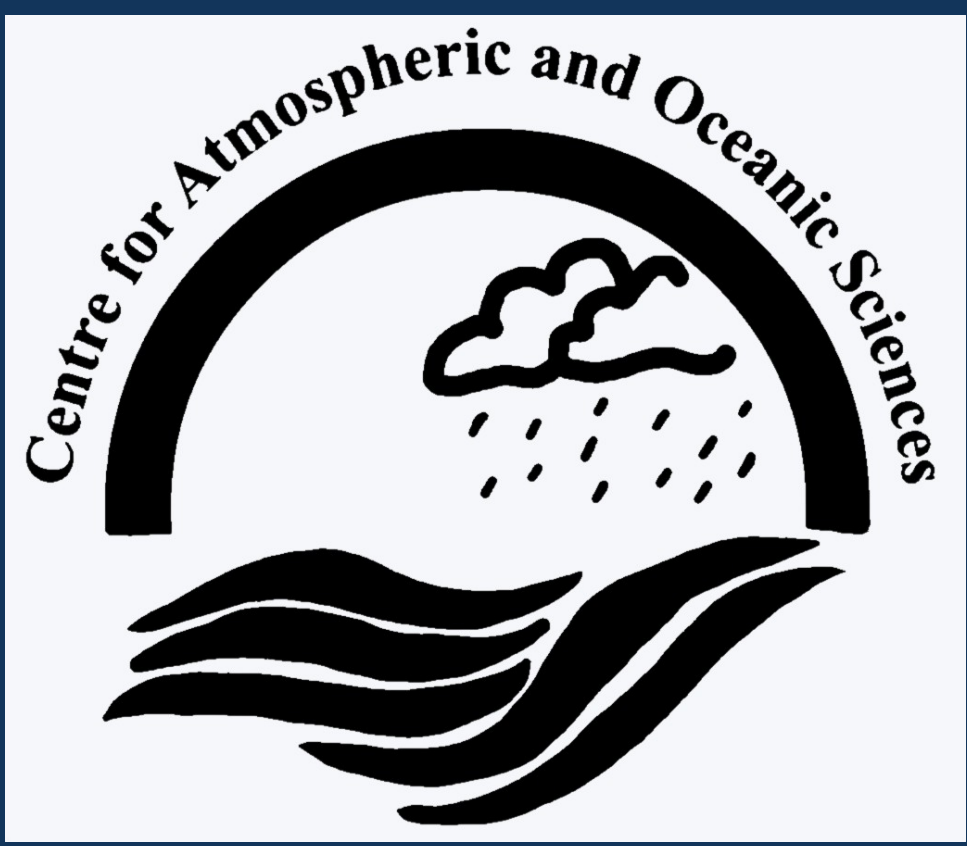




El Niño Impact on Indian Monsoon: A Tale of Two Outcomes

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THE PUZZLE

- The Indian Summer Monsoon (June–September, JJAS) brings ~70% of India's annual rainfall, a seasonal deficit of more than 10% is a drought.
- El Niño is generally associated with deficient Indian summer monsoon rainfall (ISMR). But the relationship is not deterministic.
- Only about half of El Niño years since 1900 brought drought to India.

Why do similar El Niño conditions give drought in some years but near-normal rain in others?

We compare **El Niño + Drought (EN + Dr, 13 yrs)** with **El Niño + No Drought (EN + NDr, 10 yrs)**, focusing on subseasonal rainfall evolution, large-scale circulation and thermodynamics.

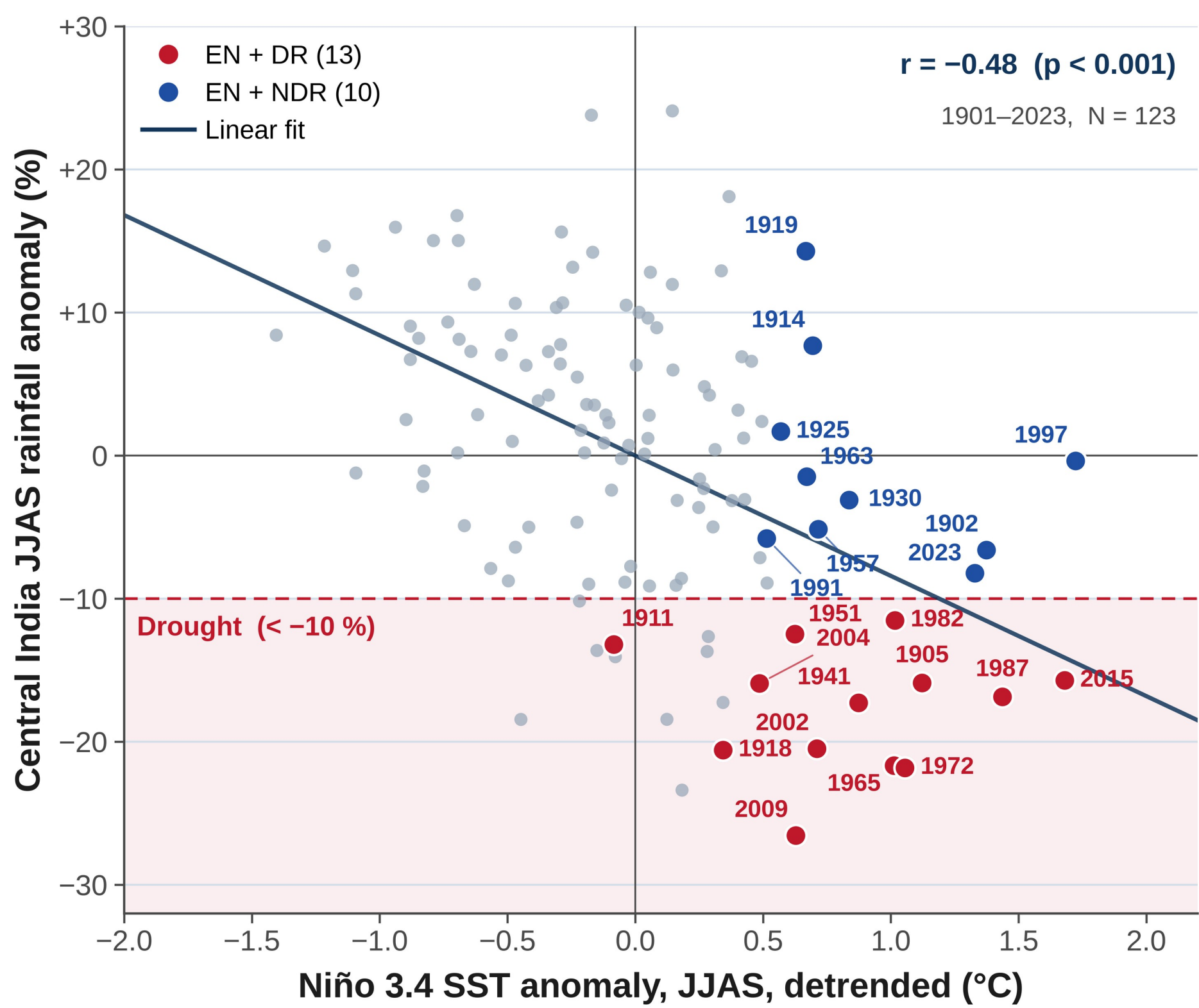


Figure 1. JJAS central India rainfall anomaly (%) versus detrended Niño 3.4 SST anomaly (°C), 1901–2023.

DATA & APPROACH

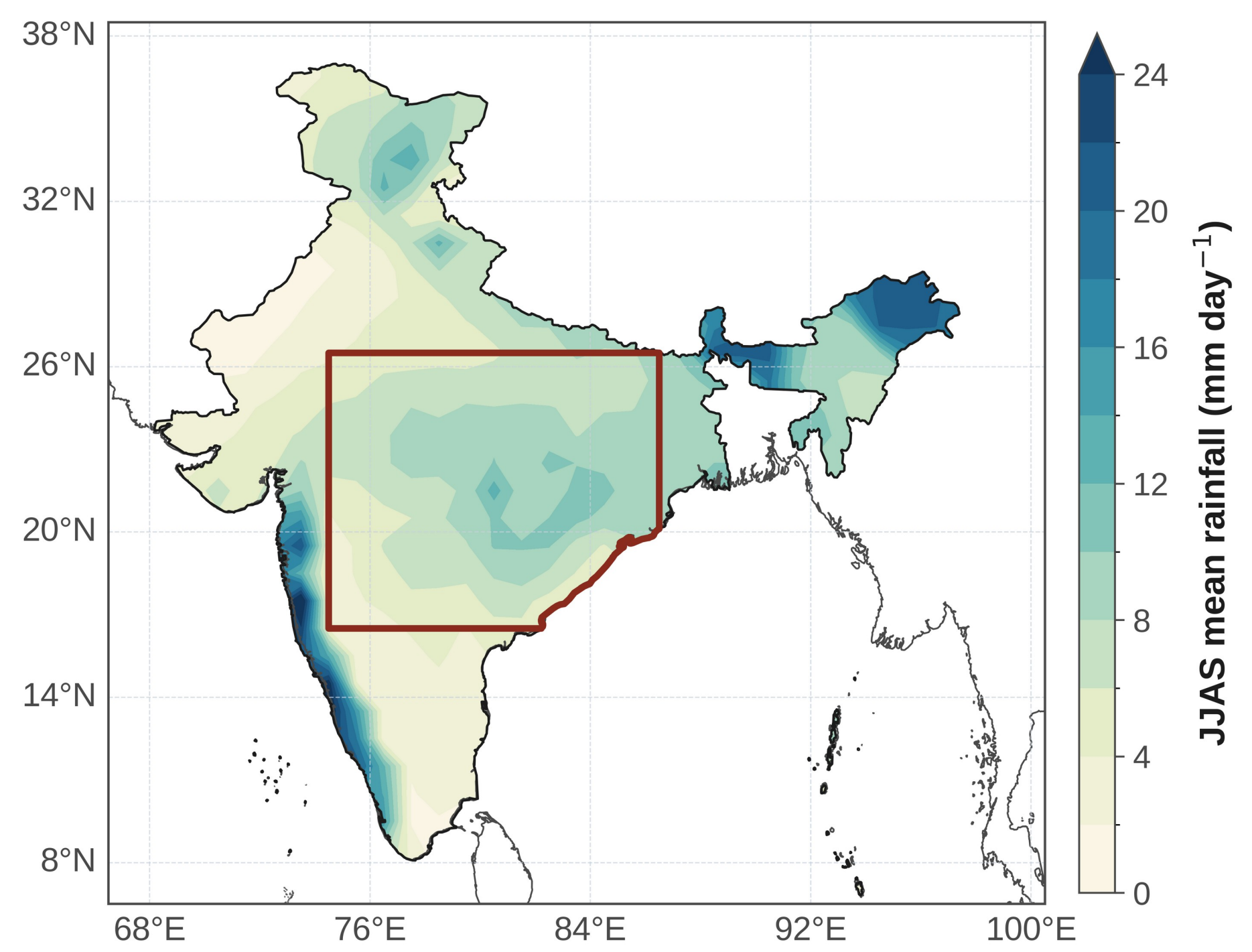


Figure 2. JJAS mean rainfall (mm day⁻¹), 1901–2023 (IMD 1° gridded data). The maroon box marks the central India study region (16.5–26.5°N, 74.5–86.5°E).

Rainfall

- IMD gridded rainfall: 1° × 1°, daily, 1901–2023 (Indian land only)

Sea surface temperature and pressure

- Hadley Centre Sea Ice and SST (HadISST1): 1° × 1°, monthly, 1870–2023

- Hadley Centre Sea Level Pressure (HadSLP2): 5° × 5°, monthly, 1901–2012

Atmospheric circulation

- ERA5 reanalysis: 1° × 1°, daily, 1940–2023

El Niño years are classified based on all-India JJAS rainfall, with drought defined as a deficit >10%.

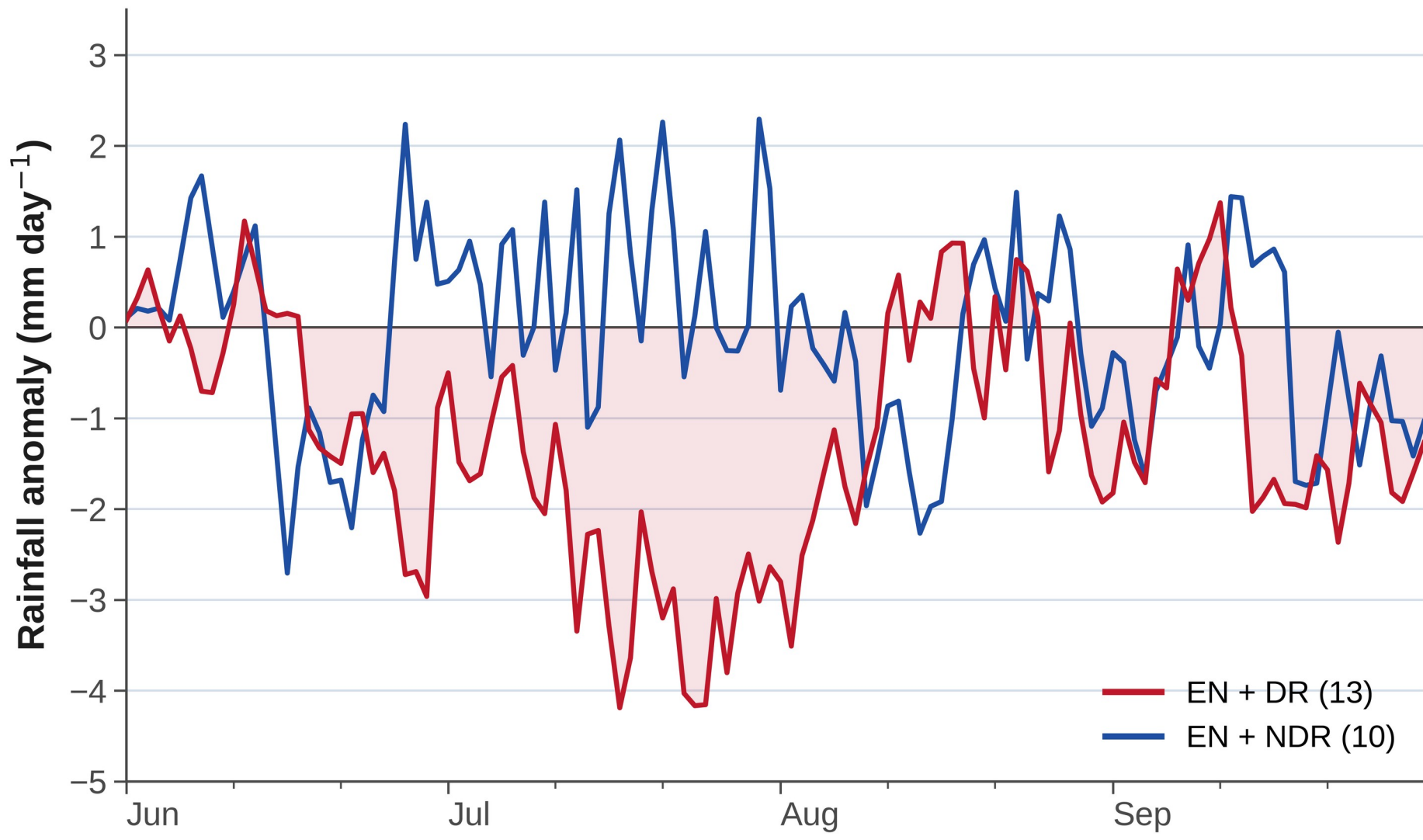


Figure 3. JJAS composite daily rainfall anomaly (mm day⁻¹) over central India (box in Fig. 2) for EN + Dr (red, 13 years) and EN + NDr (blue, 10 years), relative to the 1901–2023 daily climatology.

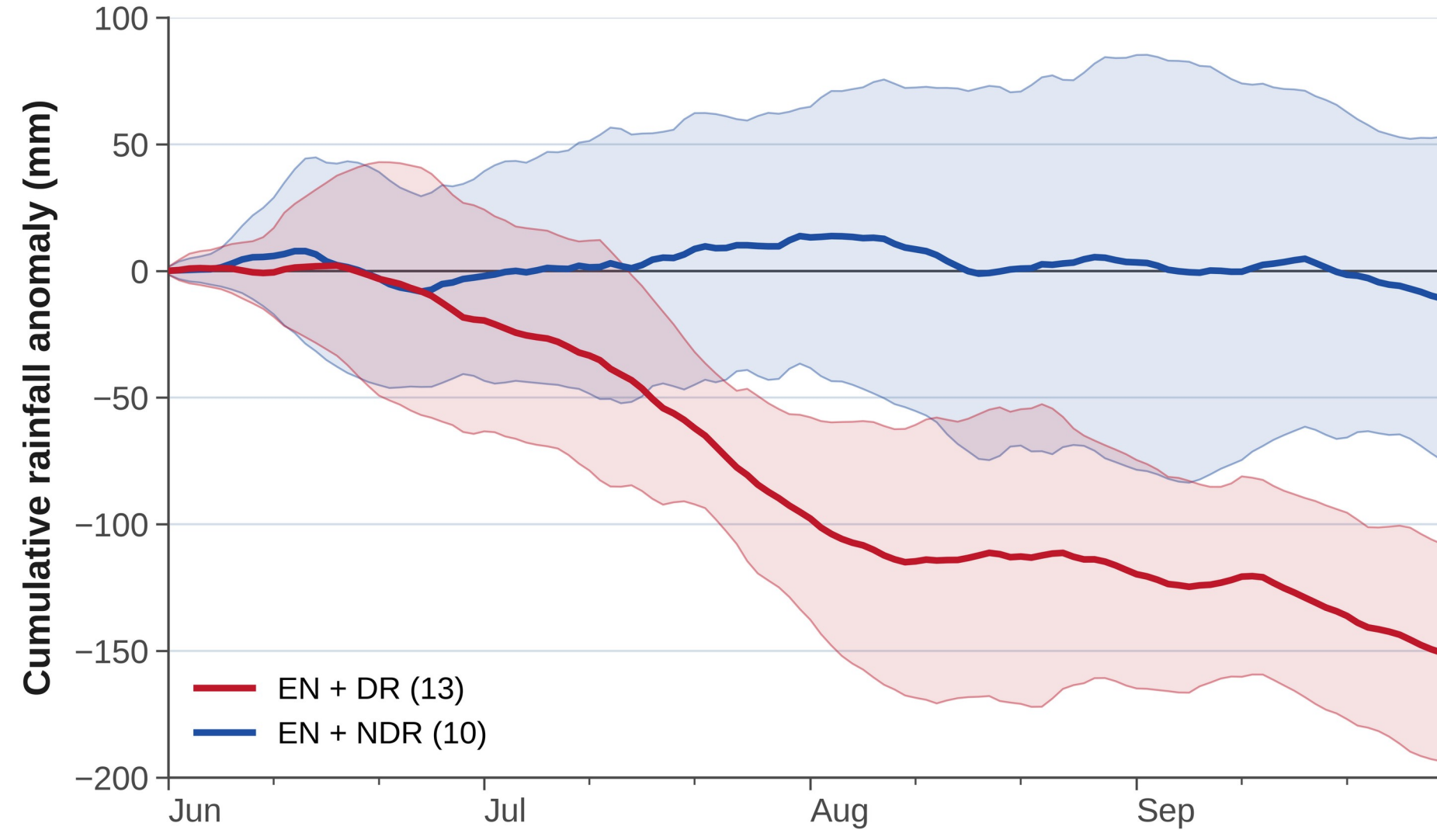


Figure 4. Composite cumulative JJAS rainfall anomaly (mm) over central India for EN + Dr (red) and EN + NDr (blue); shading shows ±1 SD across years. Anomalies are relative to the 1901–2023 daily climatology.

EN + Dr: Rainfall deficits emerge by late June and intensify through July.

EN + NDr: Deficits alternate with normal or excess rainfall, keeping the seasonal total near normal.

From mid-August: The two groups become increasingly similar; EN + NDr resembles a delayed EN + Dr response, as El Niño influence strengthens late in the season.

MONSOON RESPONSE

Deficit onset (Day 0): The 15-day running-mean rainfall anomaly first falls below -1σ . Atmospheric fields are composited in 3-day windows from Day -10 to Day $+10$.

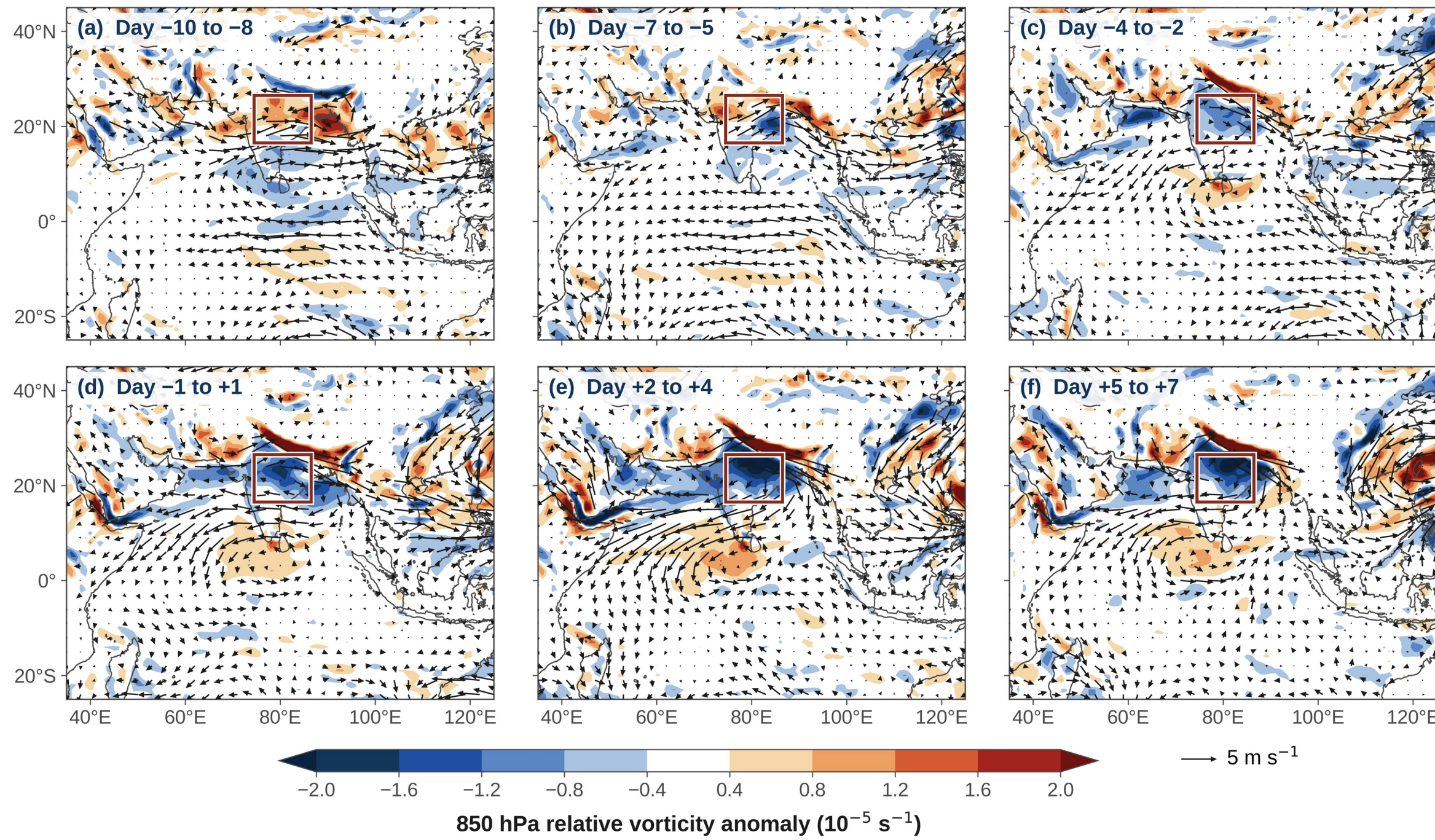


Figure 5. Composite 850 hPa relative vorticity anomaly (shading, 10^{-5} s^{-1}) and wind anomaly (vectors) for 3-day means relative to rainfall-deficit days in EN + Dr years (13 events, 1940–2023; ERA5).

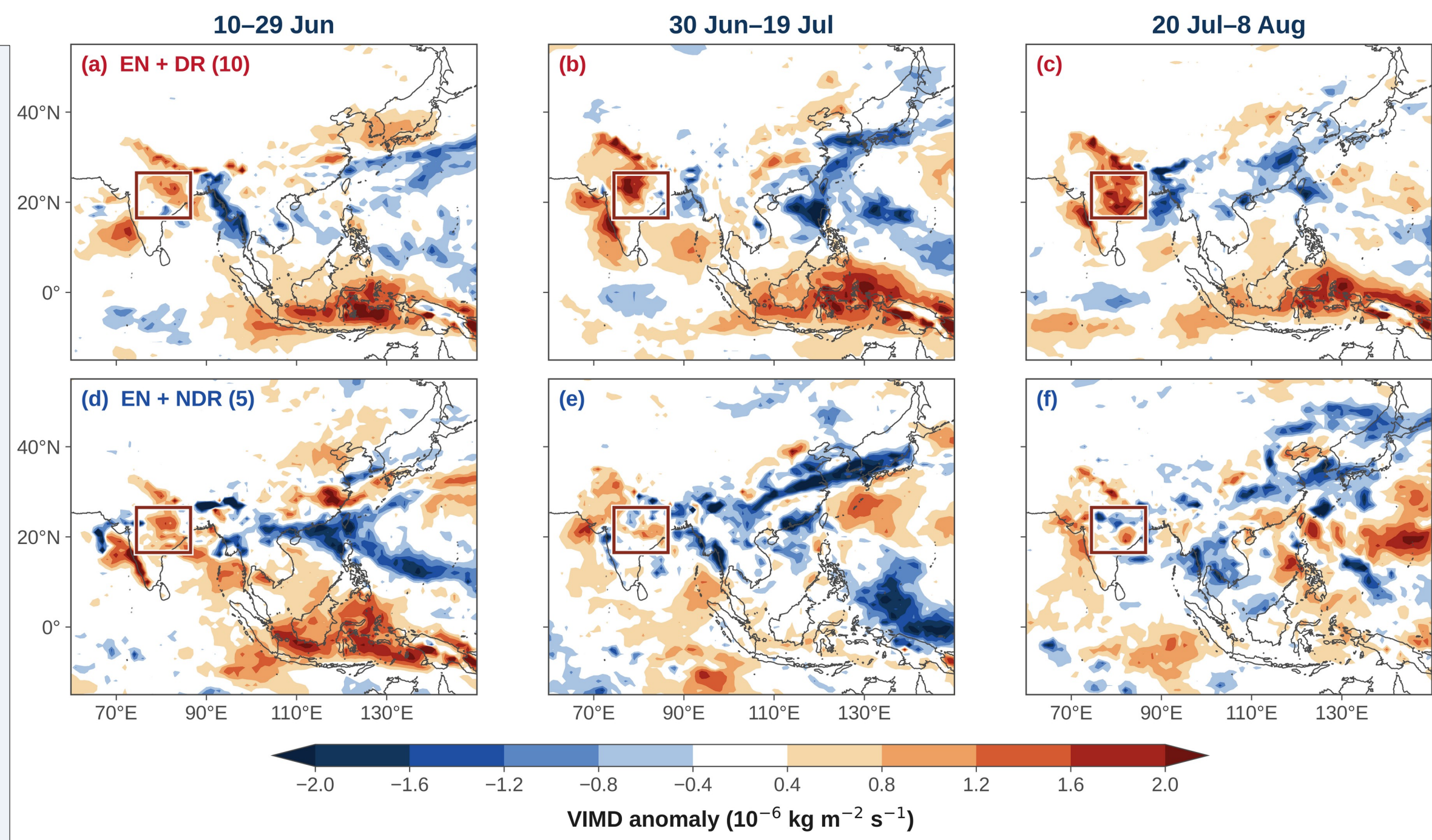
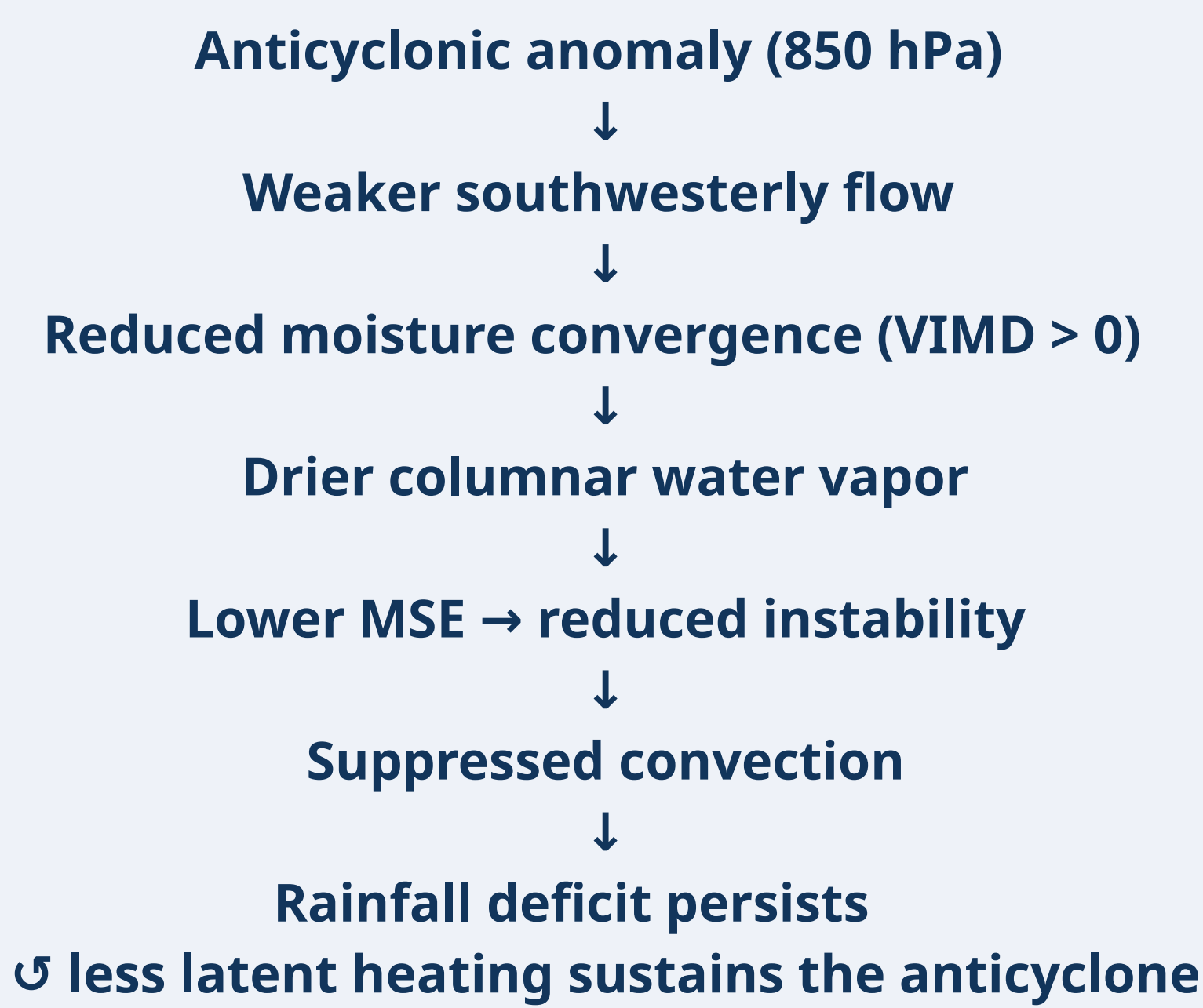


Figure 6. Composite VIMD anomaly for EN + Dr (top) and EN + NDr (bottom) years over three 20-day windows, relative to the 1940–2023 climatology (ERA5).

SUMMARY

- Not every El Niño brings drought. By season end, EN + Dr years have a cumulative deficit of about -150 mm , against about -10 mm in EN + NDr years.**
- A persistent low-level anticyclone marks EN + Dr years. It weakens moisture supply, lowering column MSE and suppressing convection.**
- EN + NDr years lack this signal. They show no sustained moisture divergence over central India.**

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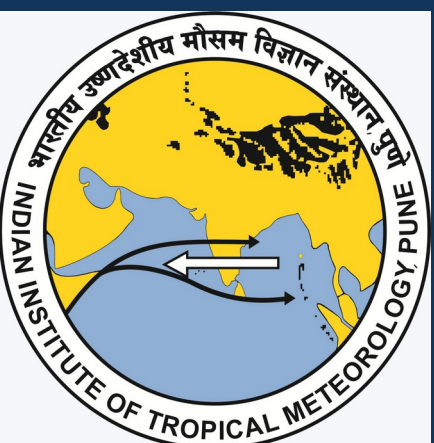
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